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Nốt

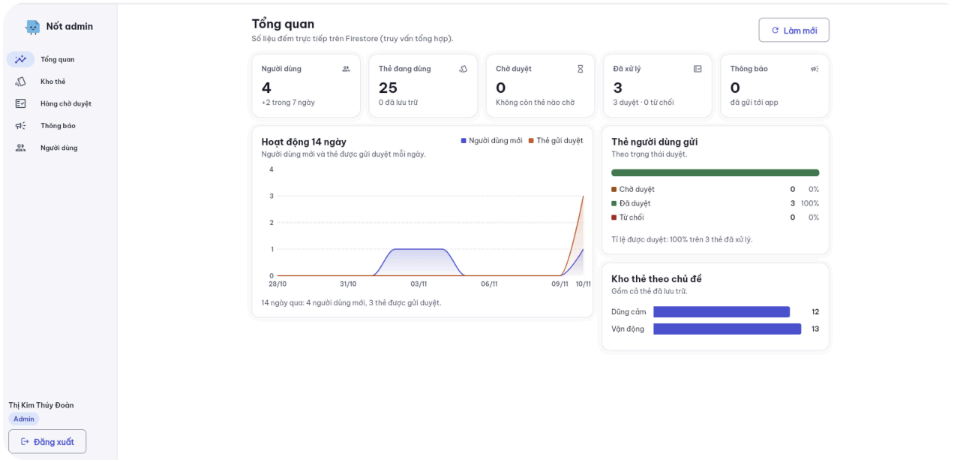
One positive note every day, now with staff, AI and a shared catalog

Đoàn Thị Kim Thúy · SE190992

Phùng Thế Quân · SE193519

Phan Đức Duy · SE193591

Phan Hữu Phước · SE161951





AGENDA · FOUR PARTS

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Problem → Design → Demo → Proof

01

Problem

Why build it, and what is it?

Why build it?

- 01 Overview
- 02 Context
- 03 Problem

What is it?

- 04 Solution
- 05 Actors
- 06 Features

02

Design

How is the whole system built?

- 07 Business flow
- 08 State machine
- 09 Architecture
- 10 Stack
- 11 UI/UX
- 12 Sequence
- 13 Classes
- 14–15 Database
- 16 Traceability

03

Demo

How do the workflows run for real?

- 17 Demo plan
- 18 WF-01 Nốt AI
- 18b WF-01 storyboard
- 19 WF-02 Publish a card
- 19b WF-02 storyboard
- 20 WF-03 Announce
- 20b WF-03 storyboard

04

Proof

What can we prove?

- 21 Testing
- 21A Test cases
- 21B Defects
- 21C Evidence wall
- 22 Achievements
- 23 Limitations
- 24 Lessons learned
- 25 Conclusion

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01

Problem

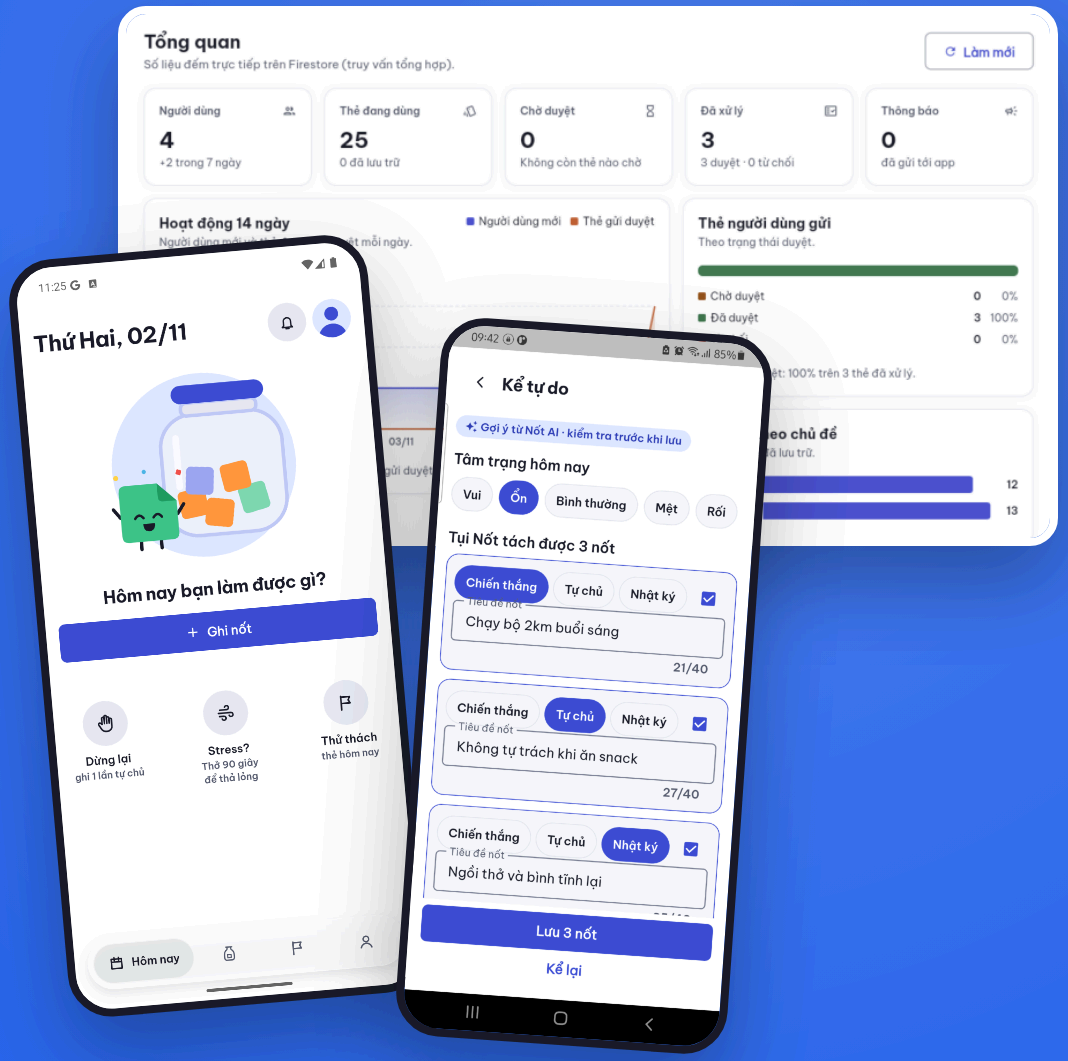
Why build it, and what is it?

Why build it?

01 Overview | 02 Context | 03 Problem

What is it?

04 Solution | 05 Actors | 06 Features



PROBLEM · 01

Overview

PROJECT

Nốt

Small-wins journal · Android app + web admin · one Firebase project

> For **students and young workers** who end a long day feeling they got nothing done: Nốt makes **one small win visible in under 30 seconds** and keeps **fresh small challenges** coming.

USERS

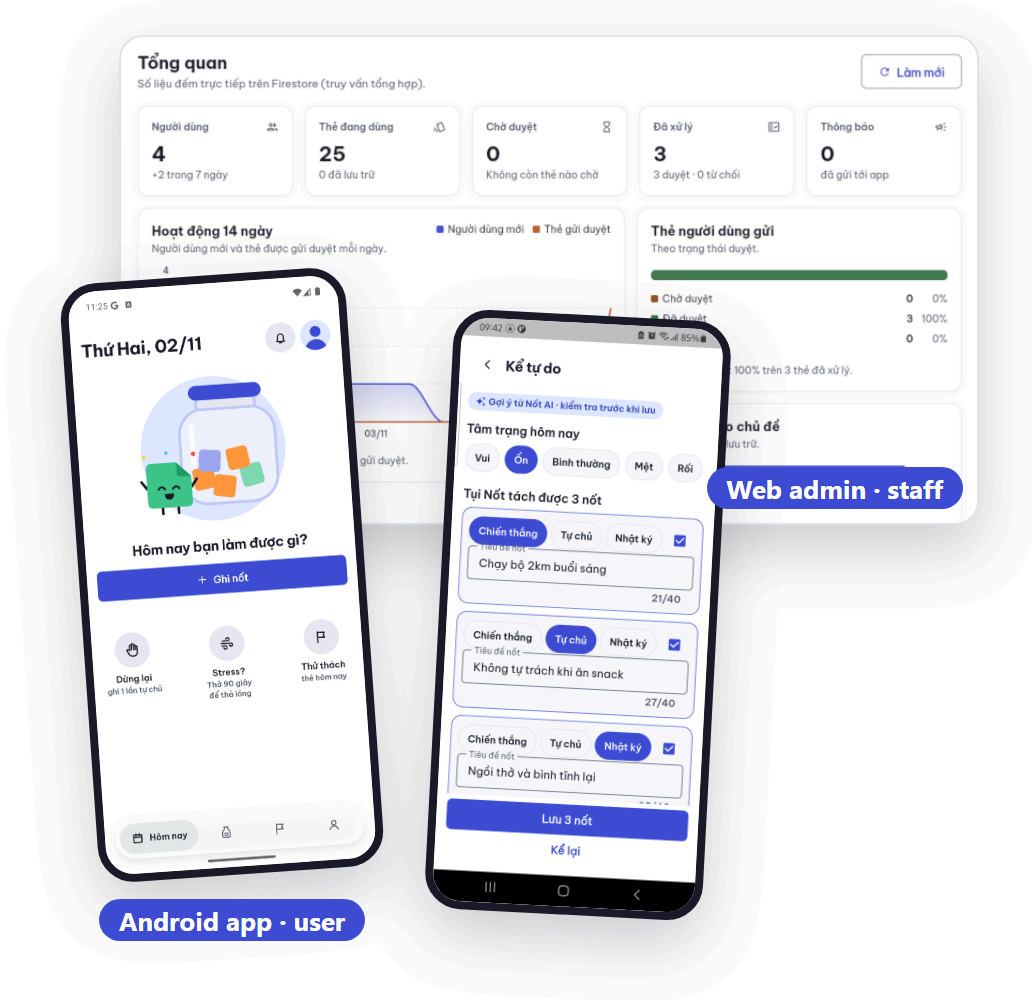
user
logs the day on the Android app

staff
reviews and creates cards, posts news on the web admin

admin
everything staff can, plus changes roles

CORE VALUE

- One win in 30 s, typed or sorted by Gemini
- New challenge cards without an app update
- Staff news in every inbox in the same minute



PROBLEM · 02

Context

Final-year students and young workers reach the evening with little energy left.

ENVIRONMENT

University life and first jobs

long days of classes, internships, part-time work; the evening is the only free time

WHO IS INVOLVED



Minh, 22 · final-year student, tired, wants to feel the day counted

Nobody curates small, fresh challenges for people like Minh

CURRENT TOOLS

To-do app, phone notes, paper diary

plus self-help tips scattered across social media

CURRENT PROCESS

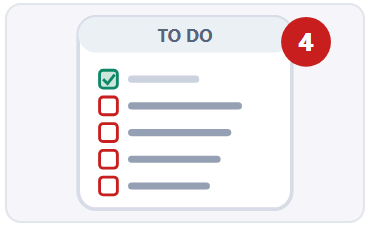
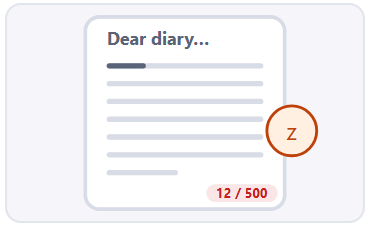
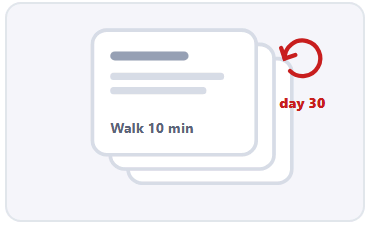
1. **Day ends** — home late, low energy
2. **Checks to-do** — sees only what is left
3. **Tries a diary** — blank page, gives up
4. **Same goals** — last week's list again
5. **Goes to bed** — "nothing done today"

So people need: a way to see one small win in seconds, and fresh small challenges that someone keeps up for them.

PROBLEM · 03

Problem

Most of us think it after a long day, even on days when we did plenty of small things.

PAIN POINT		CAUSE		IMPACT
	**At night we only see what is unfinished**	→	To-do lists track **what is left**; the small things we did are never written down	→ We feel the day was wasted; motivation drops and **burnout** builds up
	**Writing the day down is one more task**	→	Diary apps ask for a **long entry**, at the moment we are most tired	→ We skip it, then **stop after a few days**; the good moments are lost
	**Goals that never change stop motivating**	→	**Nobody refreshes** the challenges or tells us what is new	→ The same list every week gets boring; good habits **break off**

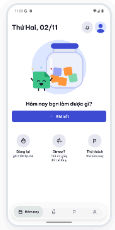
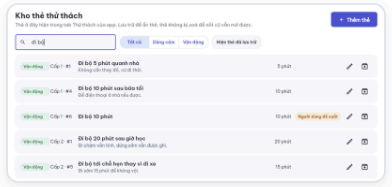
HIT HARDEST final-year students and young workers, whose evening is the only free time and the most tired one.

EVIDENCE Lab 2 persona (Minh, 22) and our own evenings as final-year students. No formal survey was run.

PROBLEM · 04

Proposed solution

An Android app for the evening, a staff web admin for content, and Gemini that sorts the day.

PAIN POINT		SOLUTION IN NỐT	VALUE
At night we only see what is unfinished	→ 	**The jar of small wins**: one mood + one win, saved in under 30 s	→ The evening ends on **what we did**, not on what is left
Writing the day down is one more task	→ 	**Nốt AI**: tell the day in one paragraph; Gemini splits it into sorted notes to confirm	→ **3 notes** from one paragraph, no form to fill
Goals that never change stop motivating	→ 	**Staff web admin**: keep the card catalog fresh, approve cards users share, announce news	→ New cards and news reach users **without an app update**

IN SCOPE WF-01 Log the day with Nốt AI · WF-02 Publish a user card · WF-03 Announce news · onboarding, weekly PDF, roles, App Check

OUT OF SCOPE Cloud Functions and Cloud Storage (paid plan) · "Hỏi Nốt" AI (bonus) · iOS build

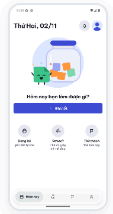
PROBLEM · 05

Main actors

The role lives in users/{uid}.role; only an admin changes it, and never their own.

User
logs every evening

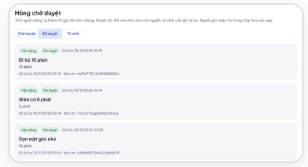
- own notes and data
- share own cards (pending only)
- read catalog, news, own inbox



Android app

Staff
content curator

- create, edit, archive cards
- approve / reject with a reason
- post announcements



Web admin

Admin
all staff rights
team lead

- everything staff can do
- change other users' roles
- never their own role



Web admin

Firebase
Auth · Firestore
App Check · Remote Config · FCM

Security Rules decide every read and write

Gemini
via Firebase AI Logic
gets the scrubbed paragraph only
returns JSON suggestions

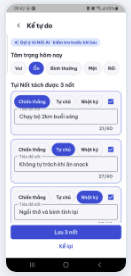
PROBLEM · 06

Main features

Each group links an actor to the pain point it solves, and names the workflow we demo live.

Log the day

User



DEMO WF-01
Nốt AI: tell the day once

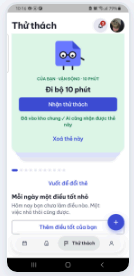
MVP
- Jar of small wins, 30 s note
- Streak and weekly card PDF

EXTRA
- Self-control and breathing

SOLVES
Only seeing what is unfinished · writing is one more task

Challenges

User Staff



DEMO WF-02
Share a card, staff approve

MVP
- Card catalog run on the web
- Onboarding + starter plan

EXTRA
- Journey climb, groups

SOLVES
Goals that never change

News to users

Staff



DEMO WF-03
Announce to every inbox

MVP
- Personal inbox messages
- Evening push reminder

EXTRA
- Discord room links

SOLVES
Nobody tells us what is new

Admin and safety

Admin



MVP
- Roles, changed by an admin only
- Dashboard
- App Check, AI on/off switch

EXTRA
- next: "Hỏi Nốt" AI, iOS

SOLVES
Keeps the other three safe

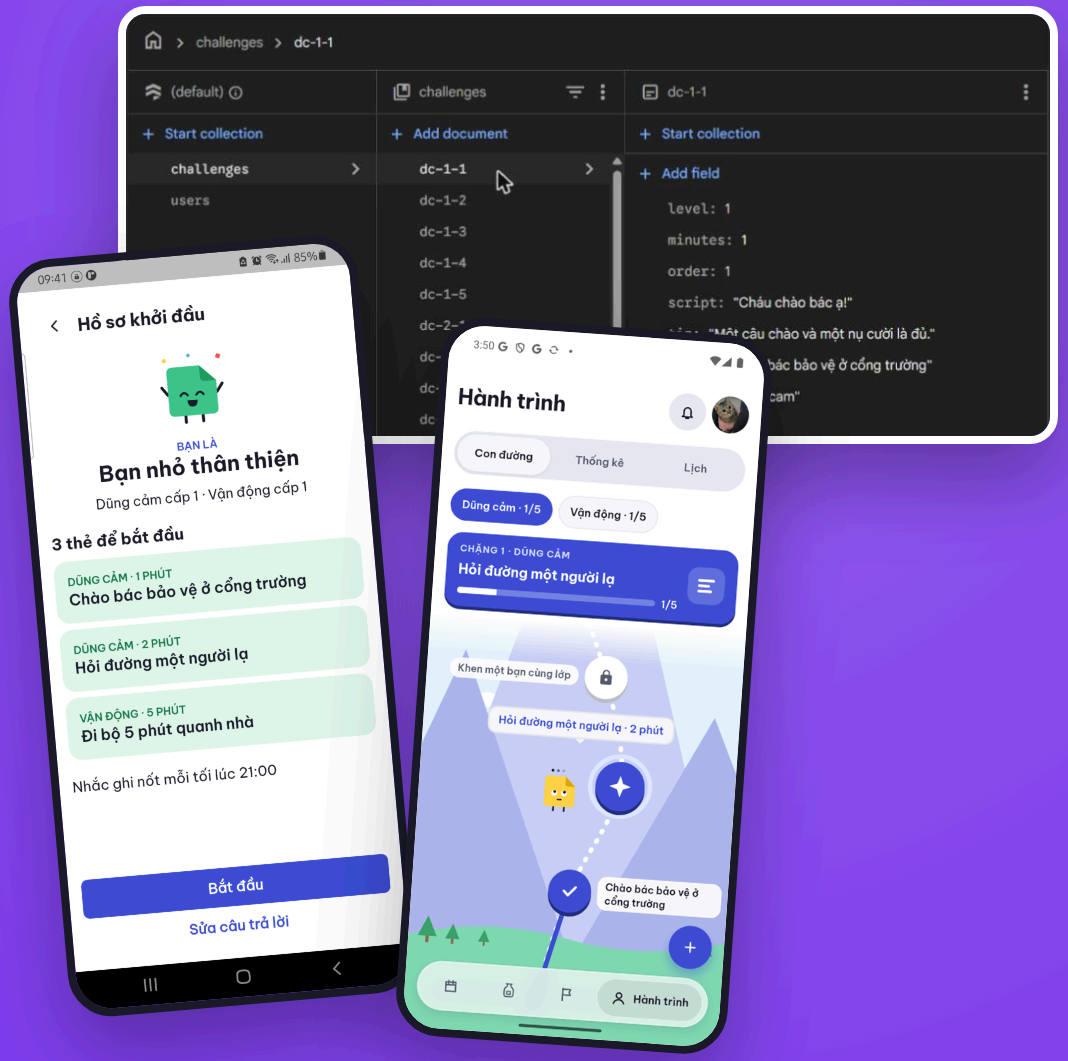
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02

Design

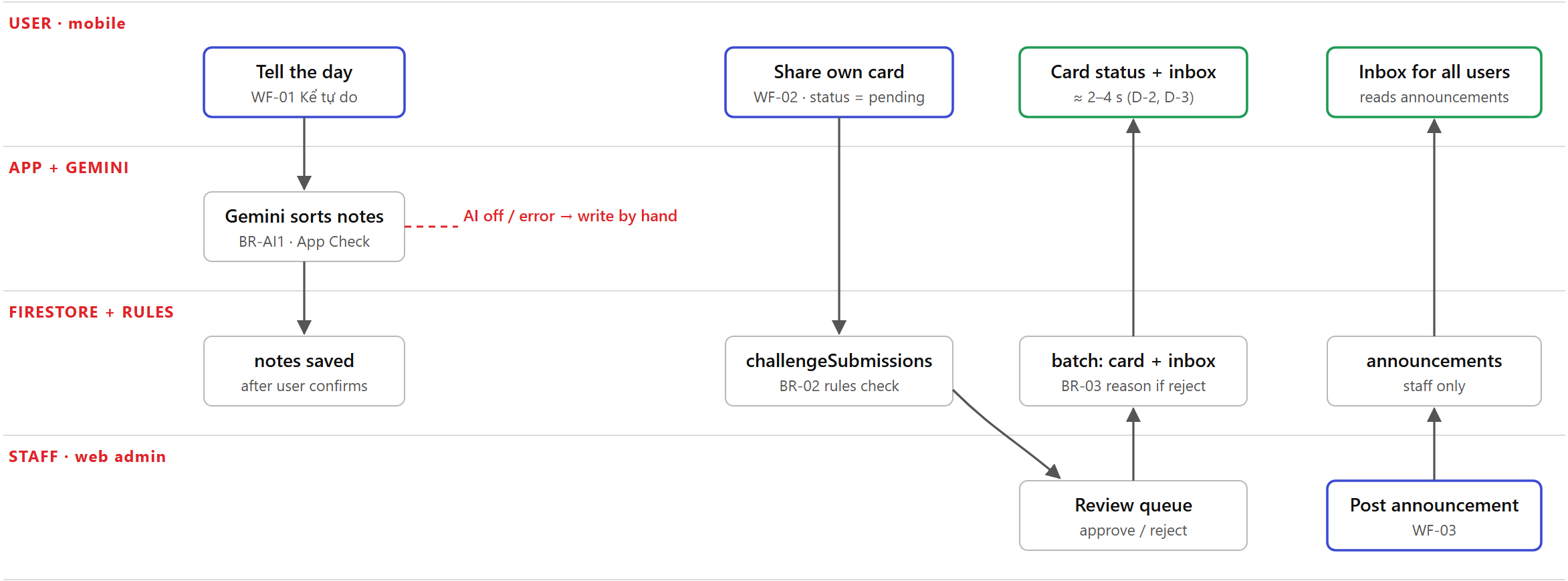
How is the whole system built?

07 Business flow | 08 State machine | 09 Architecture | 10 Stack | 11 UI/UX
12 Sequence | 13 Classes | 14–15 Database | 16 Traceability

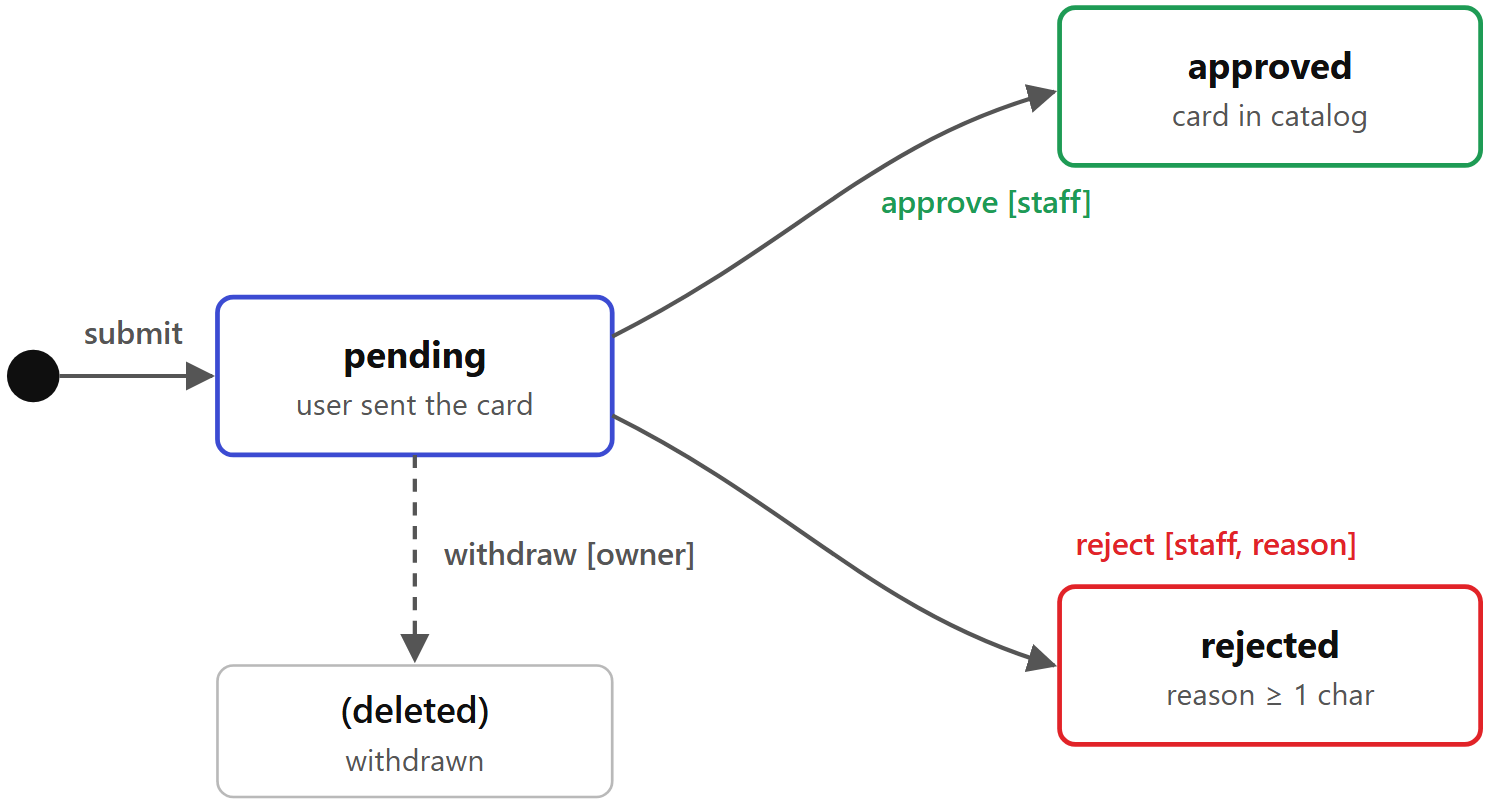


DESIGN · 07

Business flow



State machine



Forbidden (rules deny)

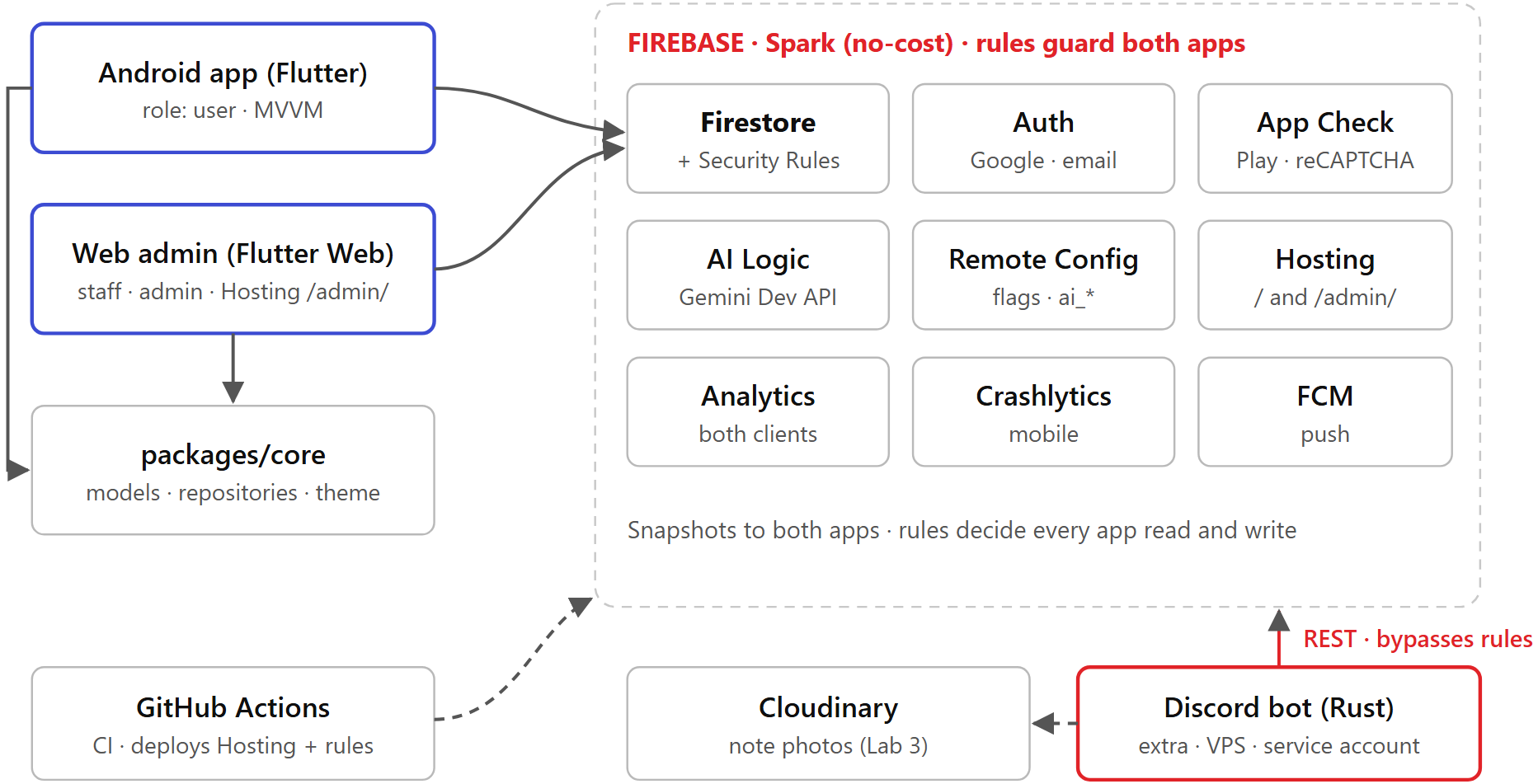
user → approved

approved → pending

reject without reason

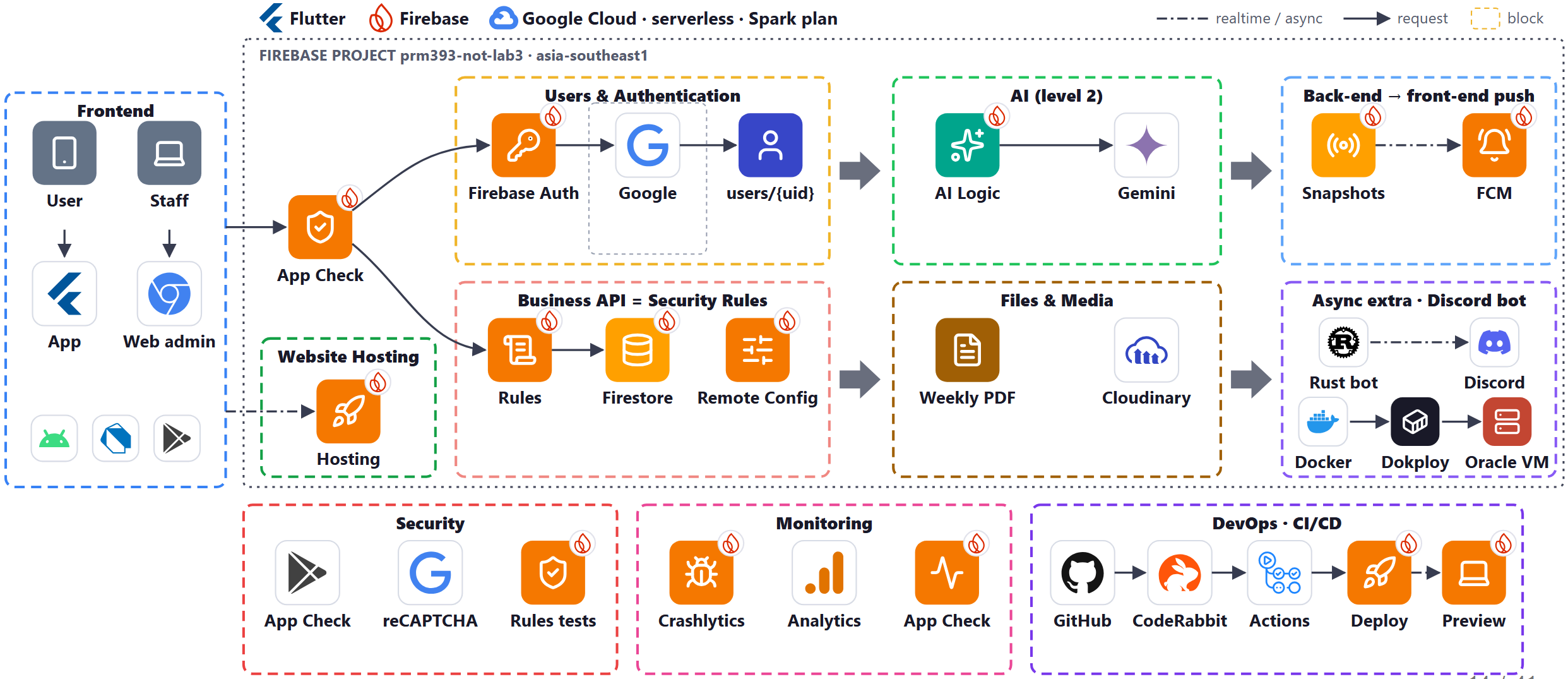
withdraw after review

System architecture



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Technology stack

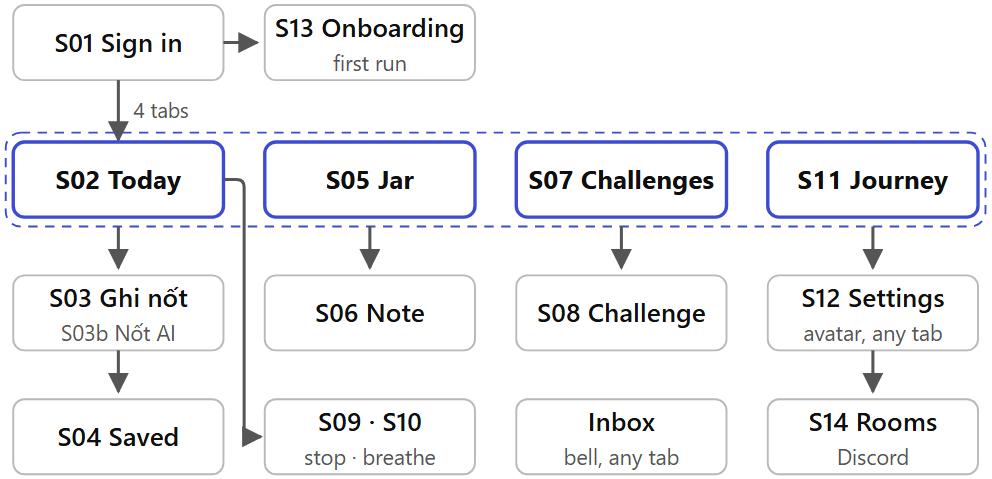


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UI/UX design

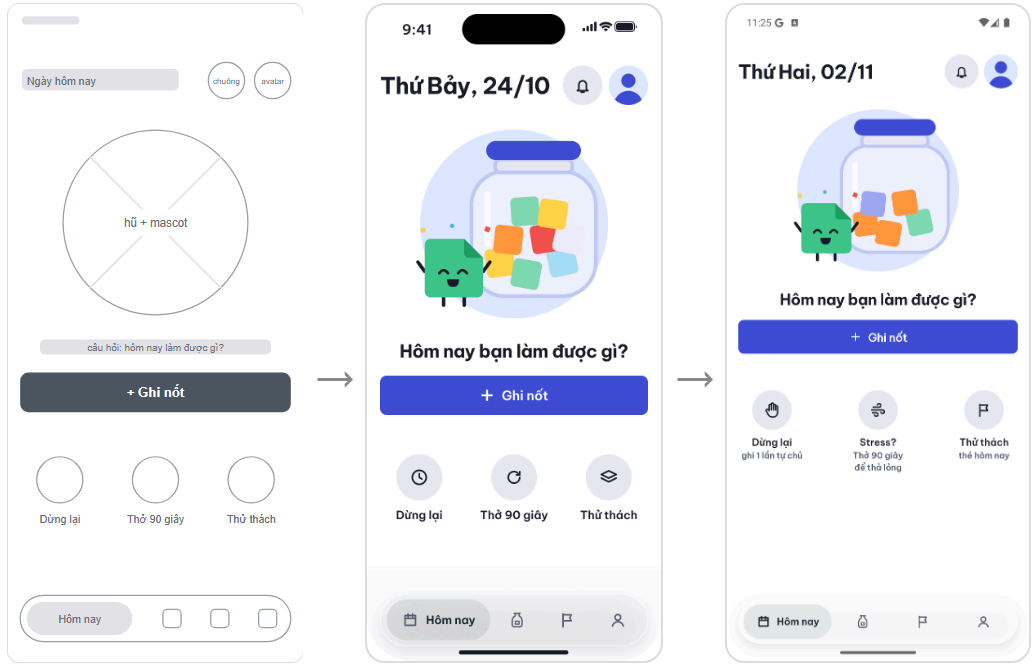
Sitemap of both clients, and how the Today screen went from wireframe to final Figma to the app.

ANDROID APP · user



WEB ADMIN · staff, admin





Wireframe
S02 Today, low-fi

Final Figma
page 03 Final UI

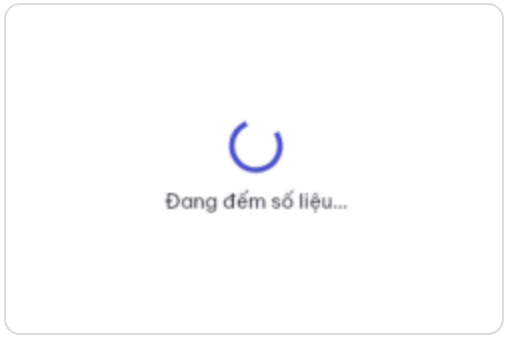
Built app
Flutter, Galaxy A70

UI states and feedback

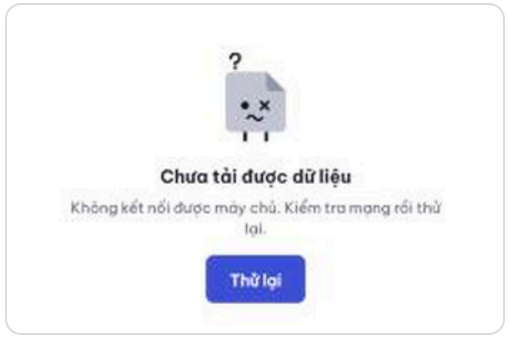
Every screen has empty, loading, error and success; three rounds of feedback changed the design.



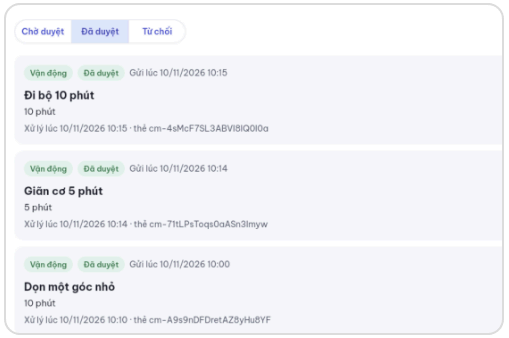
Empty · review queue



Loading · dashboard



Error · server unreachable



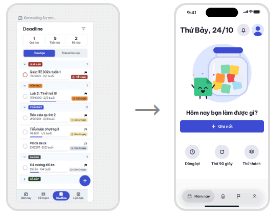
Success · card approved

FEEDBACK → CHANGE

LECTURER · LAB 2

"A deadline planner is not a real pain."

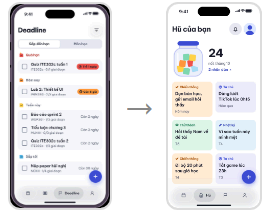
→ **Pivot to a positive-notes app; colours, font, mascot kept**



LECTURER · LAB 2

"Too much text, it feels cluttered."

→ **One big picture + one action per screen; the Jar is a grid of notes**



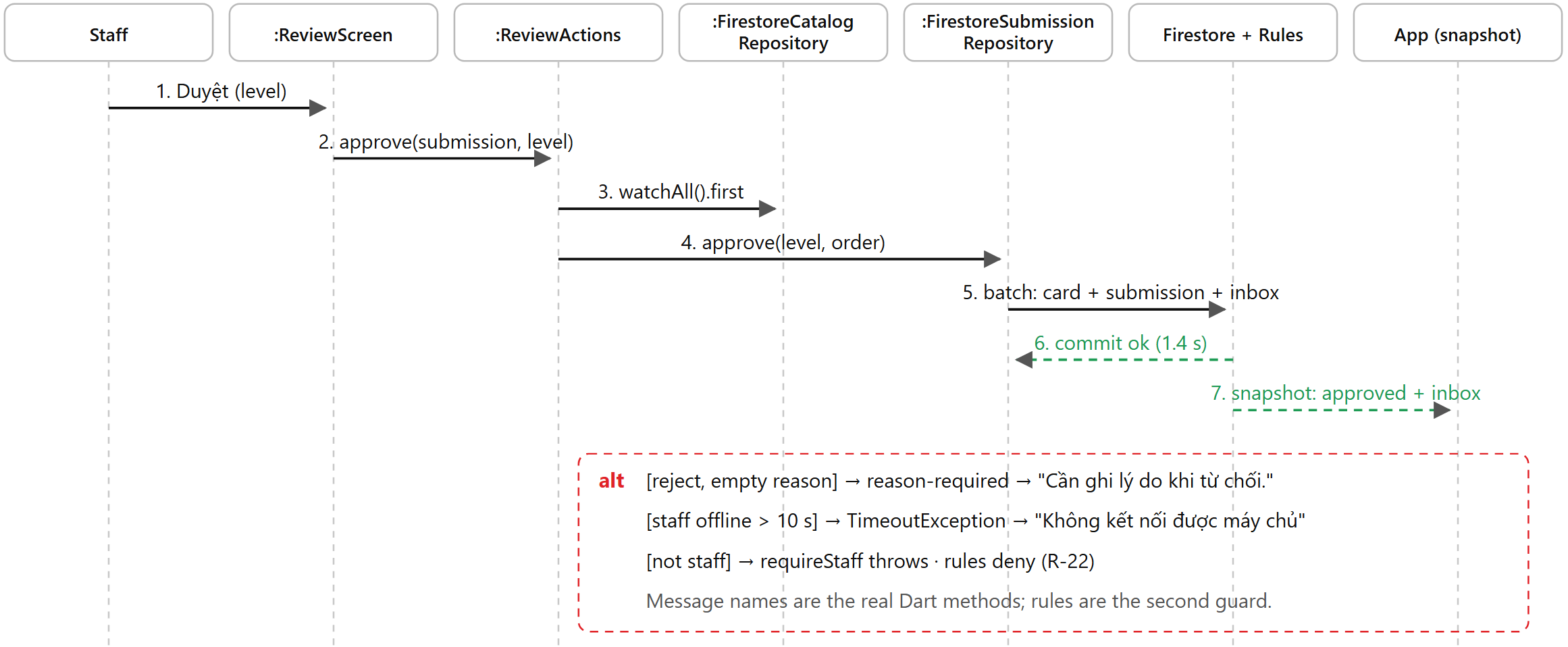
REAL PROJECT · FINAL

Errors showed only after ~3 min of retries

→ **Retries off: the error and "Thử lại" show at once**



Sequence diagram



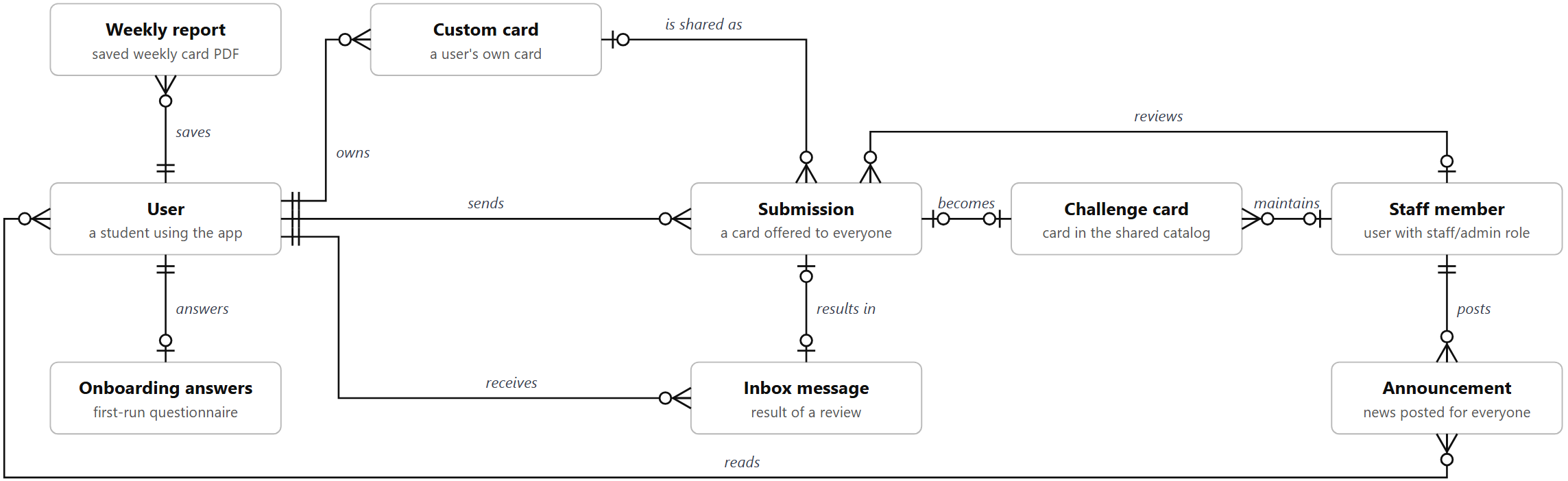
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Class design

Layer	WF-01 Nốt AI	WF-02 Share & review	WF-03 Announce
Screen	`FreeTellScreen`	`ReviewScreen`	`AnnouncementsScreen`
ViewModel	`FreeTellViewModel`	`ReviewActions`	`AnnouncementActions`
Service	`AiService`	`SubmissionRepository`	`AnnouncementRepository`
Implementation	`FirebaseAiService`	`FirestoreSubmissionRepository`	`FirestoreAnnouncementRepository`
Model	`AiSortResult`	`ChallengeSubmission`	`Announcement`

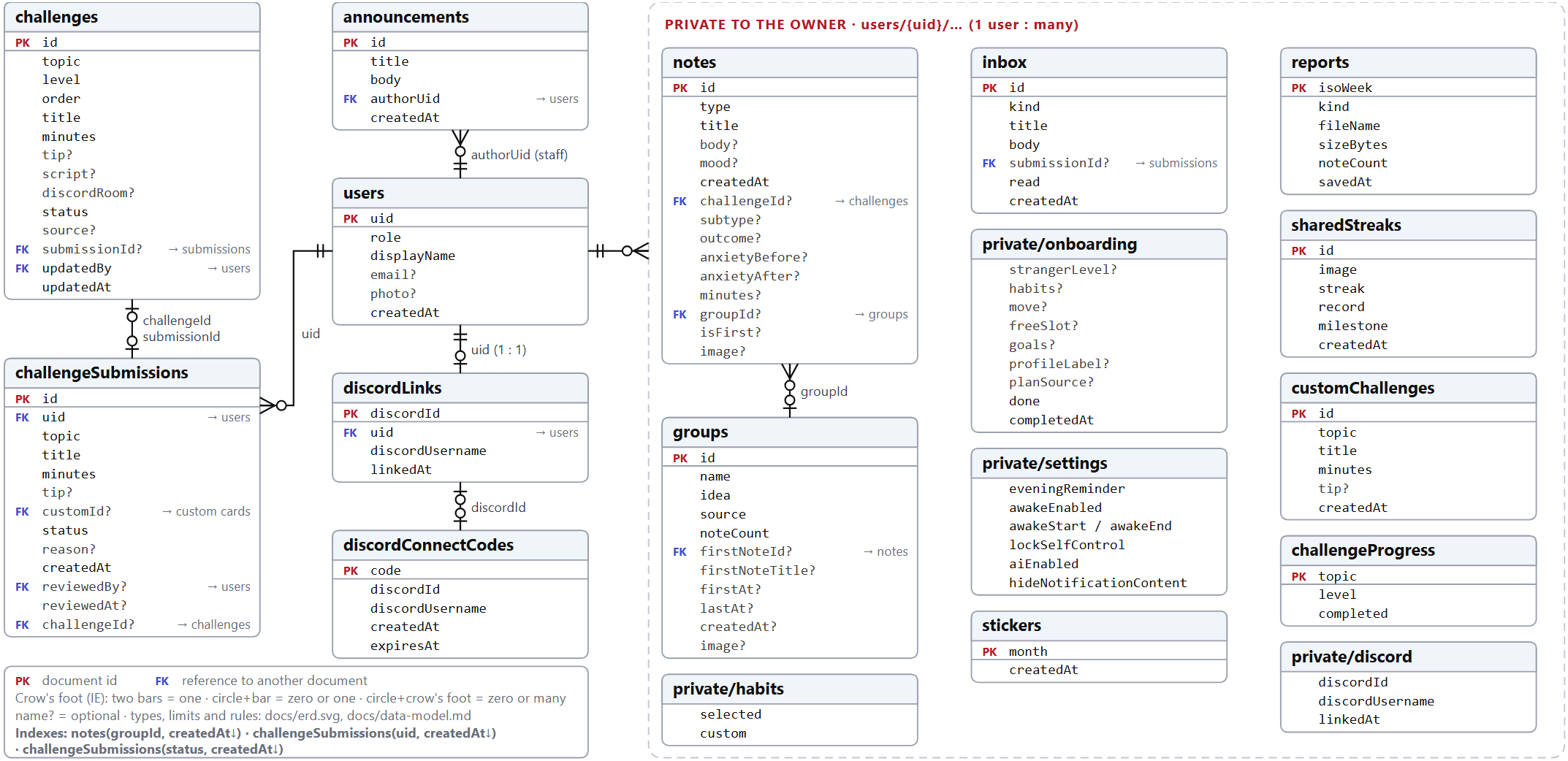
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Database design: conceptual



Crow's foot (IE) notation: two bars = exactly one · circle + bar = zero or one · circle + crow's foot = zero or many
Business level only: no attributes, keys or collection names (see the physical model on the next slide).

Database design: physical



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Traceability

Each feature traces from actor and screen to class, data, business rule and test.

Feature	Actor	Screen	Class / service	Entity	Rule	Tests
WF-01 Log the day with Nốt AI	User	S03b	FreeTellViewModel	notes	BR-AI1	free_tell_flow_test · T18, T19
WF-02 Publish a user card	User + Staff	S07 → W3	ReviewActions	challengeSubmissions	BR-02, BR-03	R-20…R-24 · cross-role 3/3 · T20
WF-03 Announce to every user	Staff	W4 → Inbox	AnnouncementActions	announcements	BR-04	R-25 · notification_inbox_test
SF-1 Weekly card PDF	User	S11b	WeeklyReportViewModel	reports	—	R-28 · T14
SF-2 Onboarding	User	S13a–g	OnboardingViewModel	private/onboarding	—	R-27 · T17
SF-3 Roles	Admin	W0, W5	sessionProvider	users	BR-05	R-01, R-26, R-29

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03

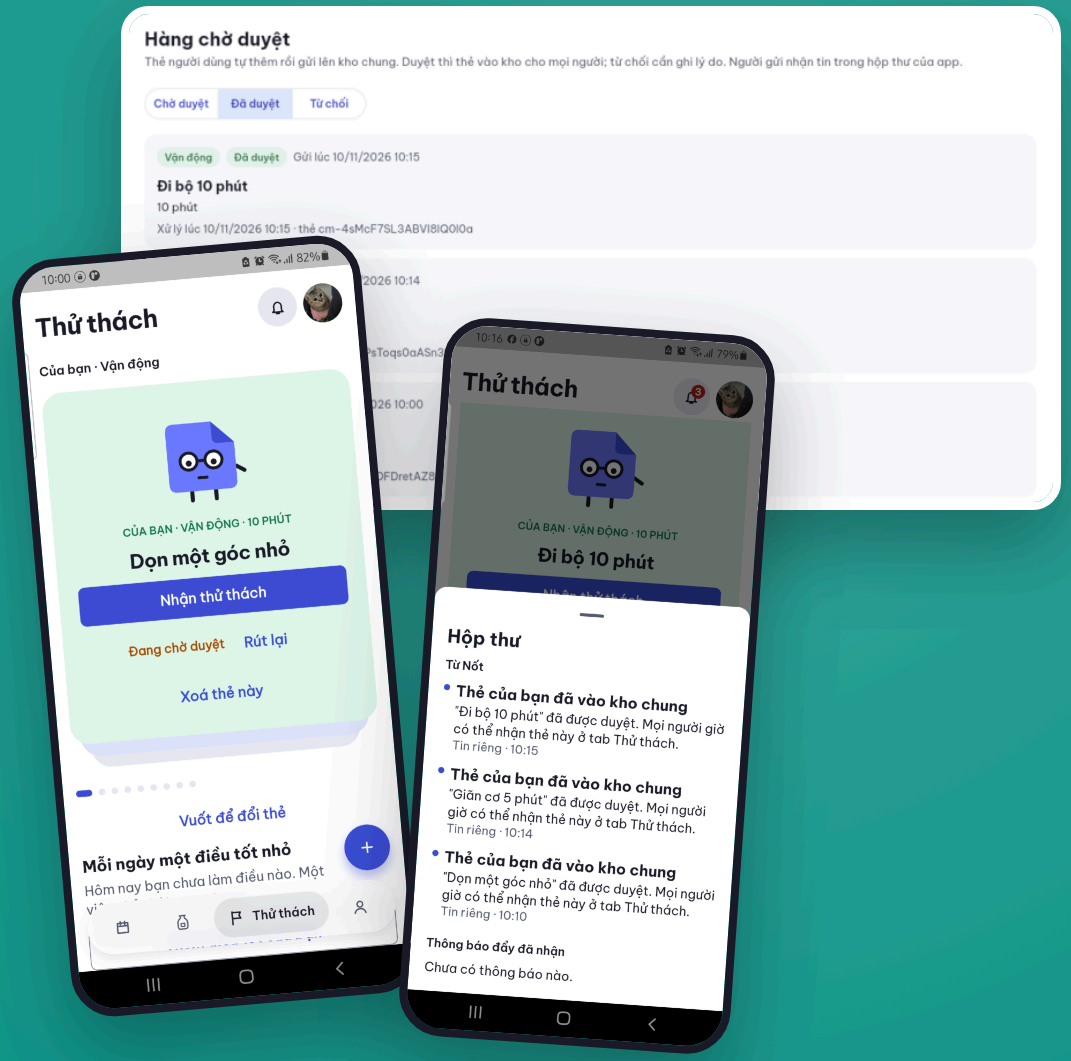
Demo

How do the workflows run for real?

17 Demo plan | 18 WF-01 Nốt AI | 18b WF-01 storyboard

19 WF-02 Publish a card | 19b WF-02 storyboard | 20 WF-03 Announce

20b WF-03 storyboard



Demo plan: one evening

Minh logs his day and shares a card; staff publish it and announce. Each result feeds the next workflow.

01 · USER
Kể tự do: one paragraph → 3 notes saved

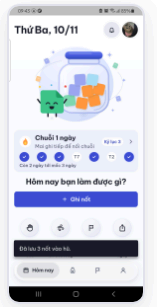
→

02 · USER → STAFF
Share "Đi bộ 10 phút" → staff approves → app shows it

→

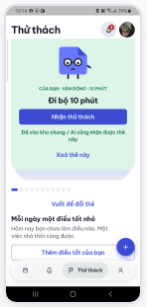
03 · STAFF → USERS
"Bạn đã vận động chưa?" → every inbox

RESULT WF-01



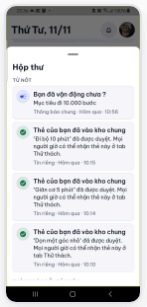
Jar 11 → 14 notes, streak starts

RESULT WF-02



Card "Người dùng đề xuất" in catalog; inbox message

RESULT WF-03



Announcement in app inbox at 10:56, same minute

EXCEPTION PATHS SHOWN LIVE

- Remote Config `ai_enabled = false` → "Nốt AI đang nghỉ", user writes by hand
- Reject without reason → blocked (BR-03) · user account on the web, staff account on the phone → rejected (BR-05)

WF-01 Log the day with Nốt AI

One paragraph becomes three editable notes.

TRIGGER · PRECONDITION

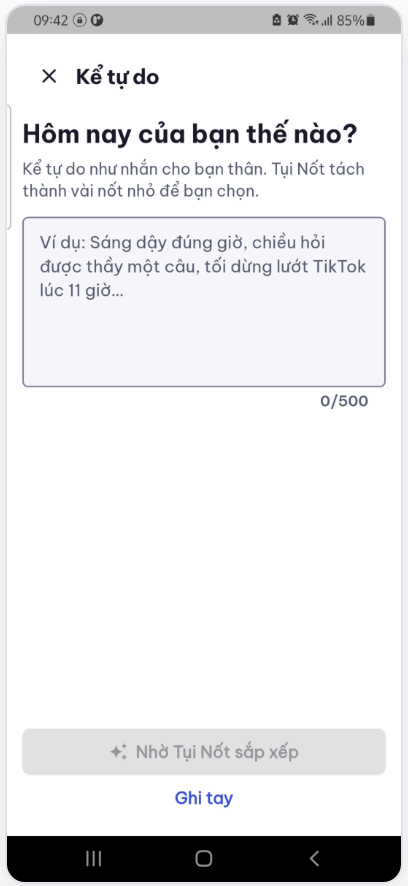
Evening, Ghi nốt → Kể tự do · role `user`, AI on (BR-01)

STEPS → RESULT

- Consent sheet on first use
- Scrubbed paragraph → Gemini → JSON notes + mood
- Labelled "Gợi ý từ Nốt AI", editable
- **Result:** chosen notes saved, jar and streak update

EXCEPTION

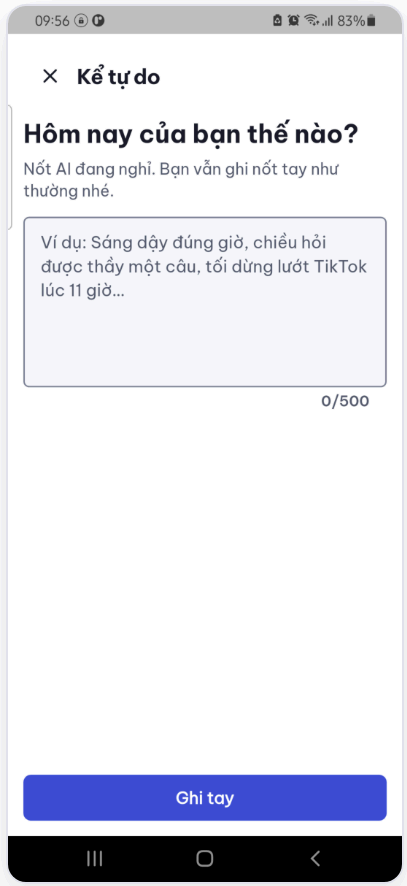
BR-01 fails (AI off, App Check, network) → "Ghi tay", paragraph kept



Consent



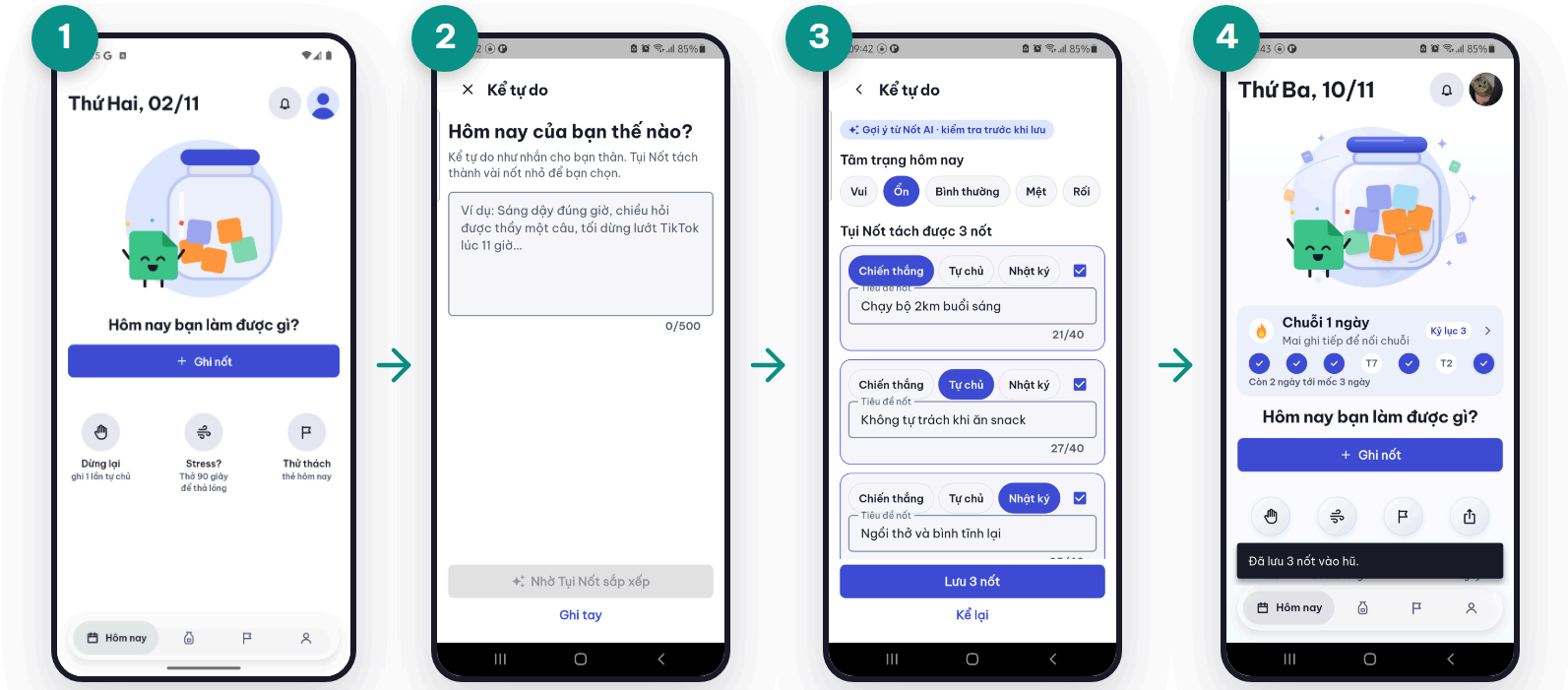
Suggestions



ai_enabled = false

WF-01 in four screens

From an empty evening to three saved notes.



Ghi nốt
evening, from Today

Consent
what is sent to Gemini

Suggestions
3 notes + mood, editable

Saved
jar 11 → 14, streak starts

WF-02 Publish a card

Staff approve on the web; the phone updates within seconds.

TRIGGER · STATE

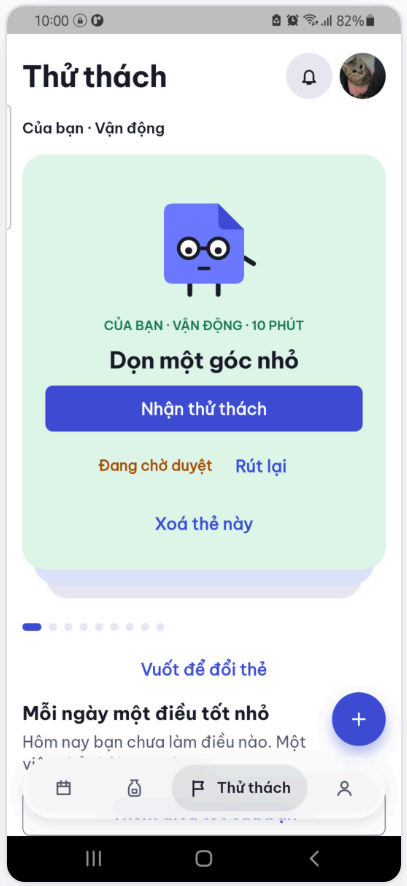
User shares own card · **pending** → **approved** / rejected

STEPS

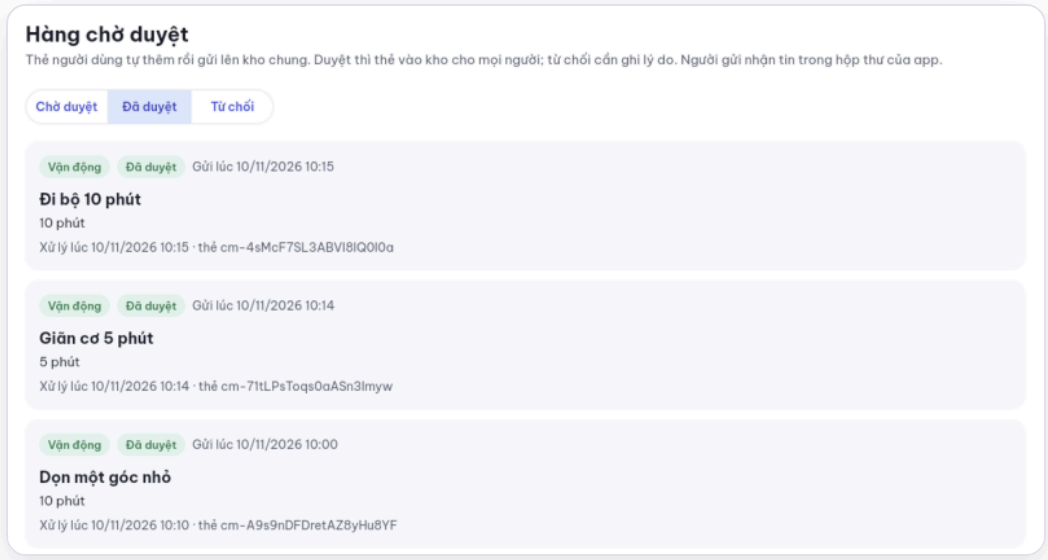
- User shares → pending
- Staff: Duyệt on the web
- Batch: card + status + inbox
- App: "Đã vào kho chung ✓"

EXCEPTION

BR-03 reason required · BR-02 user cannot approve



Đang chờ duyệt



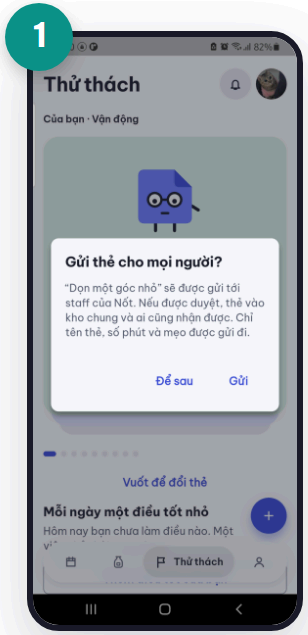
Web: approved



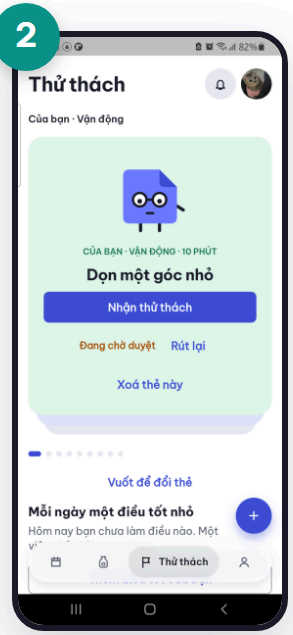
App: inbox

WF-02 in five screens

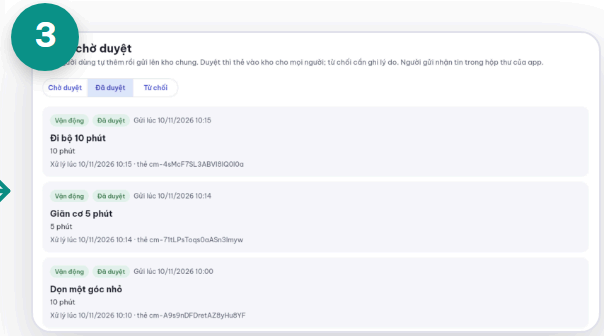
A private card becomes everyone's card.



Share
user sends own card



Pending
"Đang chờ duyệt"



Approve
staff, web admin



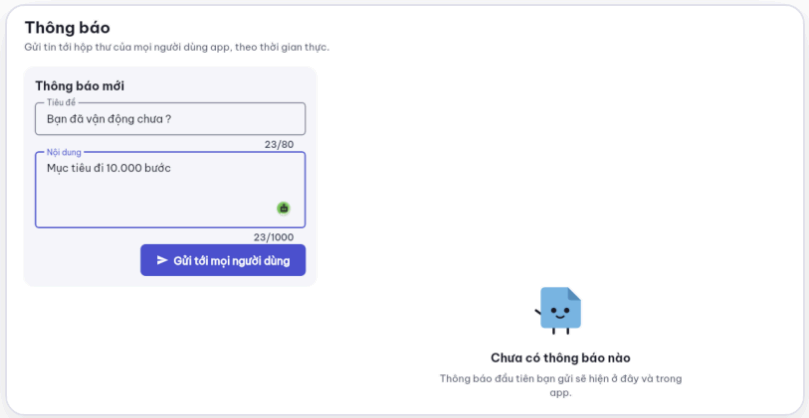
In catalog
"Đã vào kho chung ✓"



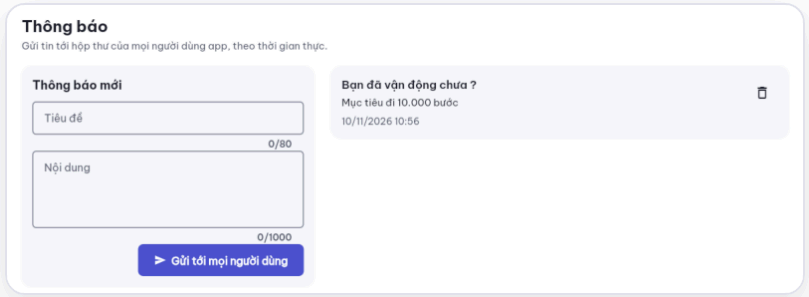
Inbox
message for the author

WF-03 Announce news to every user

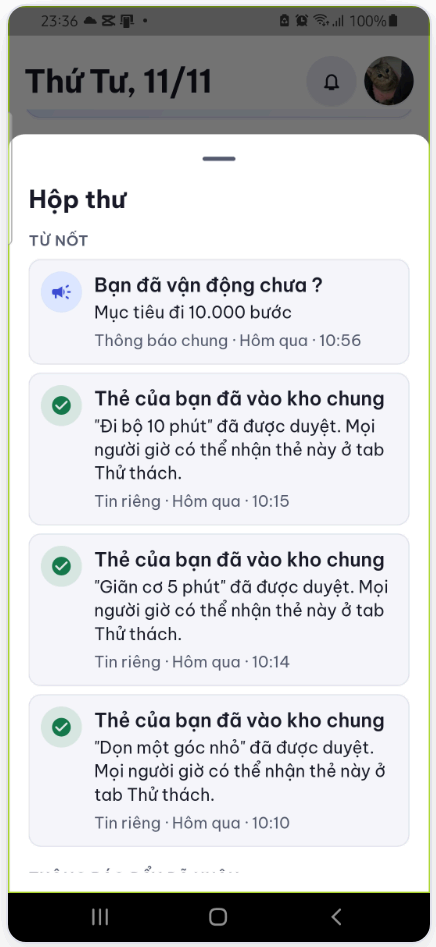
Staff post once; every inbox shows it in the same minute.



1 · Compose (list still empty)



2 · Sent at 10:56, form cleared



3 · App inbox, 10:56

TRIGGER · PRECONDITION

Staff has news for every user · signed in as staff or admin

DECISION (BR-04)

Title 1–80 · body 1–1000 · authorUid = self, in the form and in the rules

RESULT

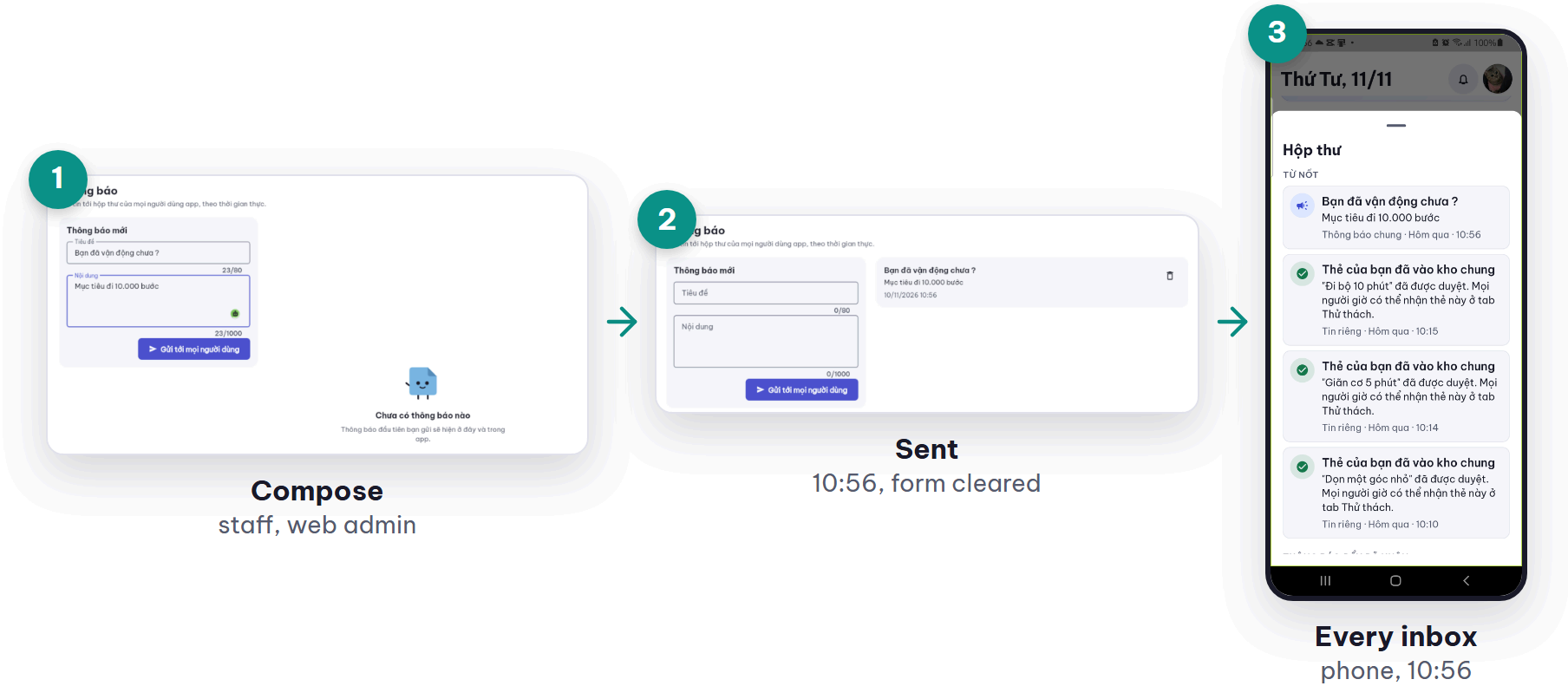
Announcement published; every user's inbox shows it in real time

EXCEPTION

Empty or too long → form error, nothing saved · user account → rejected at sign-in, rules deny (R-25)

WF-03 in three screens

One form, every inbox.



Compose
staff, web admin

Sent
10:56, form cleared

Every inbox
phone, 10:56

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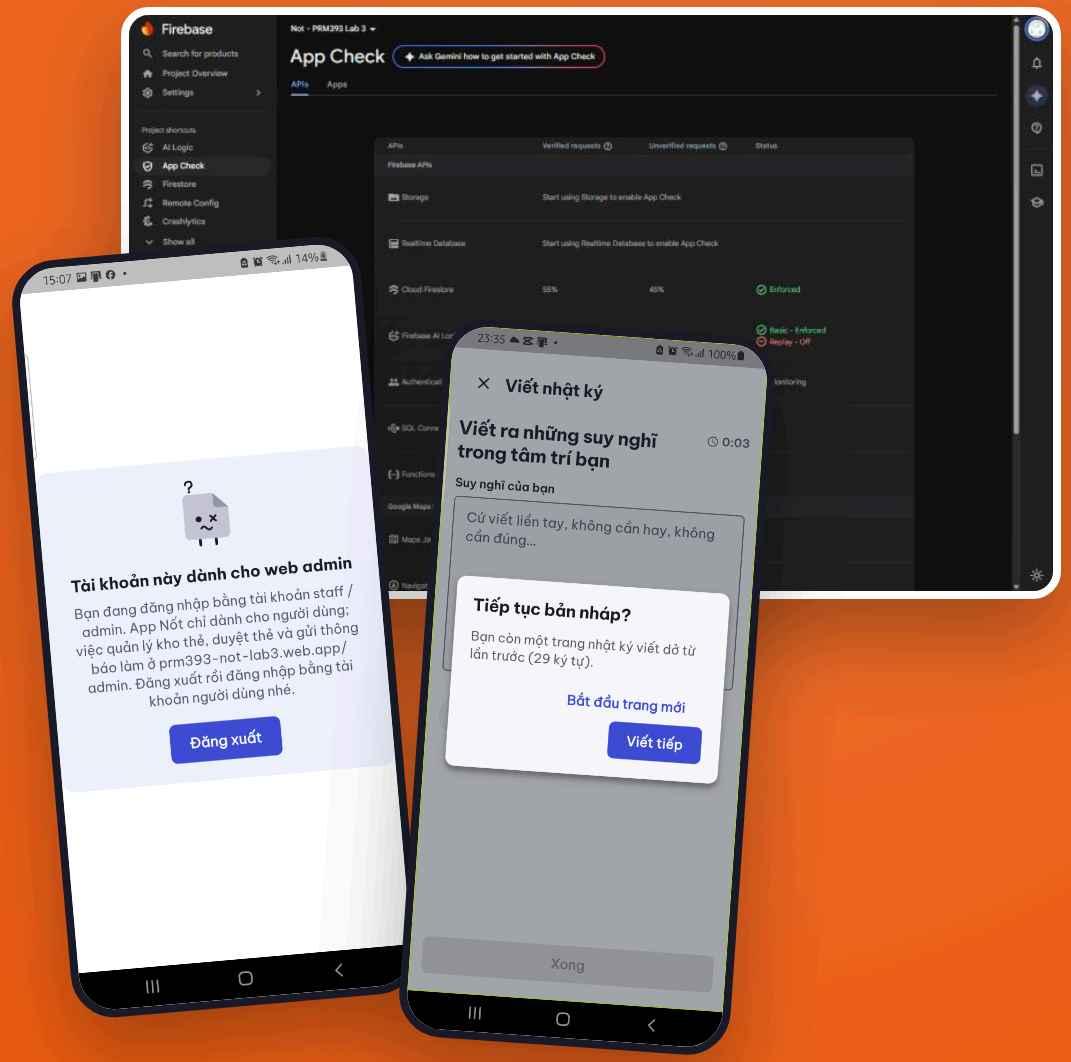
04

Proof

What can we prove?

21 Testing | 21A Test cases | 21B Defects | 21C Evidence wall

22 Achievements | 23 Limitations | 24 Lessons learned | 25 Conclusion



Testing at three levels

Rules, code and the deployed system are each tested, with evidence in the repo.

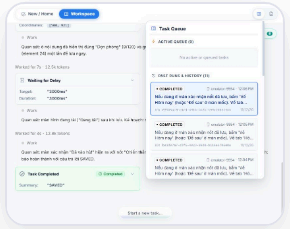
30/30
Security Rules tests

305
Flutter app tests

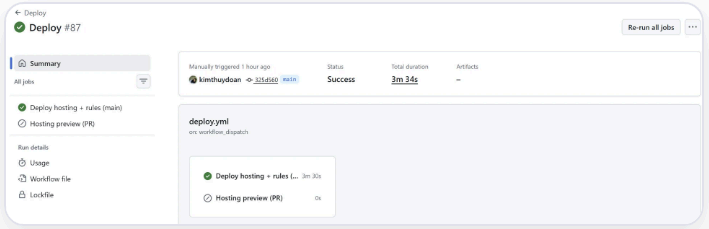
14 + 5
web admin + core tests

14/15
ARTEMIS Final runs, real phone

LEVEL	EVIDENCE	RESULT
Rules	Emulator, allowed + denied per role per collection (BR-02…BR-05)	30 / 30
Unit / widget	ViewModels, repositories, screens at 360 dp	all pass
Cross-role	3 runs on the deployed system, timed	3 / 3
Web checklist	Actions, rejection, empty, loading, error, live role change	16 / 16
Real device	Onboarding, Nốt AI, PDF, AI off, journal draft, role rejection on Galaxy A70	pass
E2E (ARTEMIS)	Lab 3: 41/42 · Final on Galaxy A70: 14/15 · emulator T17/T18/T20: 7/9 · T15/T16 not run	see batches



ARTEMIS: agent reports SAVED



GitHub Actions: deploy on main

PROOF · 21A

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Test Cases & Results

20 defined cases. Results stay attached to their device and test batch.

Batch / device	Cases and expected result	Result
Lab 3 · emulator	T01–T14, historical baseline with documented reruns	41/42
Final · Galaxy A70	T14 PDF · T17 onboarding · T18 AI · T19 AI off · T20 mobile review	14/15
Final · emulator	T17: PLAN_SHOWN, including app restart	3/3
Same emulator batch	T18: SUGGESTIONS · T20: PENDING APPROVED INBOX	2/3 each
Not run	T15 email sign-in · T16 wrong password and recovery	Not Run

Each case records preconditions, steps, expected and actual results, run outcomes and proof. A pass in an earlier batch does not replace a later failure.

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Defects & Retest

Separate application defects, automation failures and unverified reports.

Record / category	Observed issue	Decision / retest
D-01 · test harness	T17 cleanup could not find adb.exe	SDK lookup fixed; T17 3/3
D-02 · automation	T18 run 3 exceeded 480 s	Open; inspect replay and rerun
D-03 · provider	T20 controller model call exceeded 180 s	Open; rerun after provider check
APP-01…03 · application	Approval timeout, dashboard loading, stale role UI	Fixed; tests and deploy evidence
X-01…10 · exploratory	Agent-reported UI issues from the Lab 3 build	Unverified; reproduce first

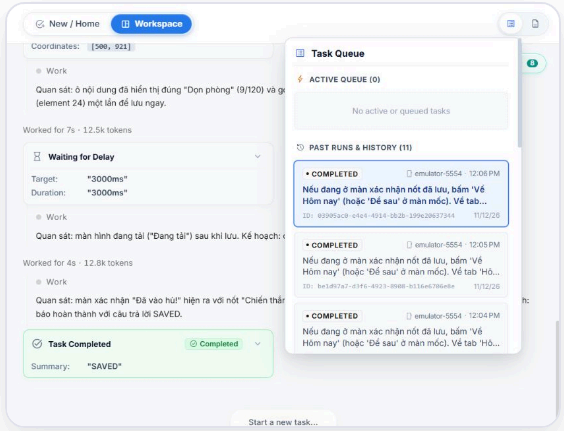
A failed test is evidence to investigate. It does not by itself prove an application defect.

Evidence wall

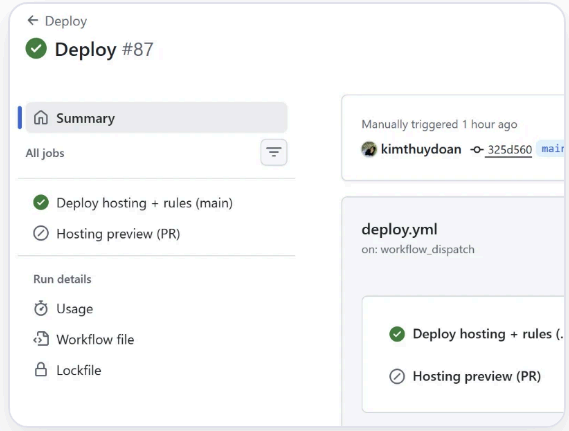
Every number on the testing slide has a screenshot or a log behind it.

```
✓ R-25 staff post announcements
✓ R-26 only an admin changes roles
✓ R-27 onboarding answers private
✓ R-28 weekly PDF metadata
✓ R-29 staff/admin cannot read private data
✓ R-30 streak card once
ℹ tests 30
ℹ pass 30
ℹ fail 0
```

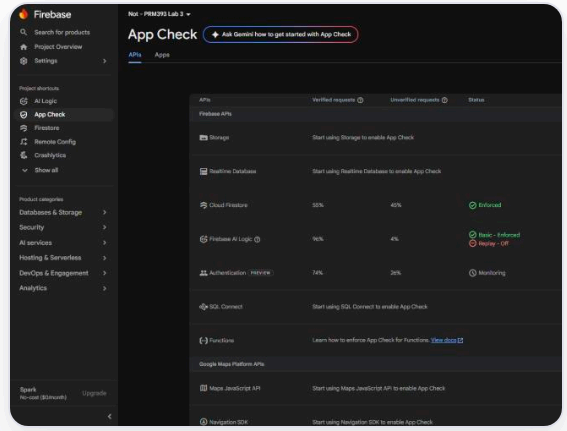
Security Rules · Emulator 30/30



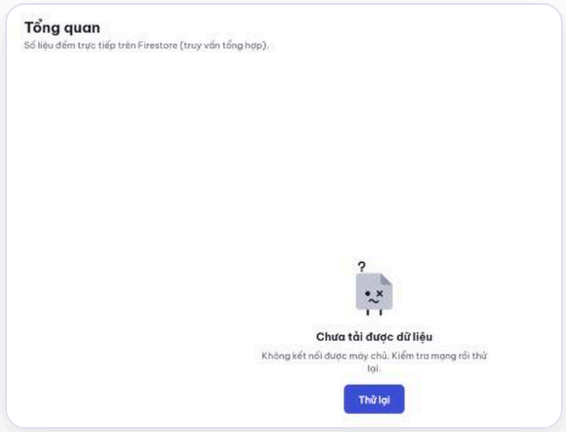
ARTEMIS on a real phone 14/15



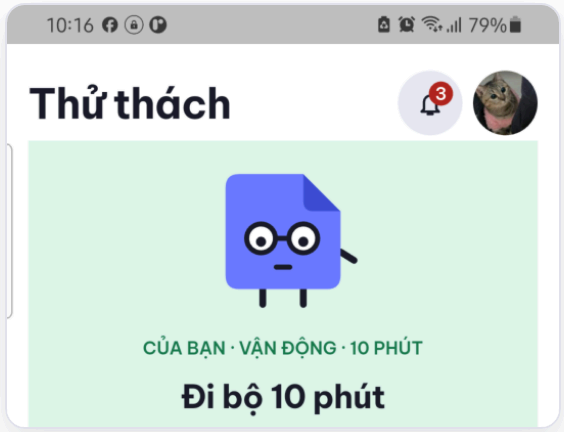
GitHub Actions deploy green



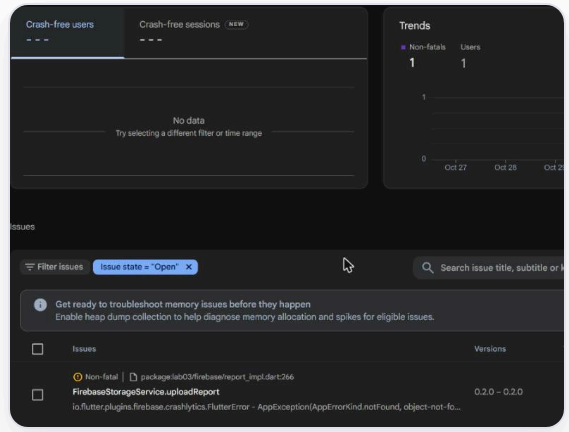
App Check enforced



Web checklist · error state 16/16



Cross-role, deployed 3/3



Crashlytics live



Analytics events live

P4 22 ACHIEVEMENTS

Achievements

Every brief requirement maps to a live feature and a piece of evidence.

BRIEF REQUIREMENT	IMPLEMENTED	EVIDENCE
Web admin: 4 screens + announcement form, deployed	Dashboard, catalog, review, announcements, users	`evidence/web-admin/` (16 items)
Role-based rules, deny by default, tested	user / staff / admin, R-01…R-30	`evidence/firestore/rules-tests.txt`
Cross-role flow, real time, documented	Share card → approve → app + inbox	`docs/cross-role-flow.md`, `evidence/cross-role/`
AI level 2 with RC, App Check, fallback	Kể tự do on Gemini, `ai_*` flags incl. prompts, enforced App Check	`docs/ai-feature.md`, ARTEMIS T18/T19
App Check on Firestore and AI Logic	**Enforced** on both · Web: reCAPTCHA Enterprise; Android: Play Integrity	`evidence/app-check/`
Lab 3 PDF saved & shared on device	pdf + share_plus, metadata in Firestore	15.4 KB file pulled from the phone
Mobile data on Firestore with inbox	Announcements + personal inbox, real time	web checklist #11

Not met: ARTEMIS T15/T16 (email sign-in) not run → slide 23

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Limitations and next steps

What is not done yet, and what it costs.

LIMITATION	IMPACT	PRIORITY	NEXT STEP
ARTEMIS T15/T16 (email sign-in) not run	Sign-out / sign-in by email has widget tests only; latest emulator T18/T20 were 2/3 each	Medium	Create a test-only account, run 3× each; investigate the timeouts
Names can reach Gemini (CR-F2)	Email and phone are scrubbed, but a friend's name is sent as typed	Medium	Consent sheet says so; add on-device name masking
Exploratory session X01: 10 UI issues not rechecked	Possible small layout bugs on rarely used screens	Low	Re-run X01 on the Final build, fix or close each
D-1 took 22 s on an idle phone	Real time not guaranteed after long idle	Low	Measure more runs; re-listen on resume
App Check on Authentication only monitored	26% of sign-in requests are unverified; Firestore and AI Logic are enforced	Medium	Enforce once all traffic is verified

PROOF · 24

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Lessons learned

Three bugs appeared only on the real project; three habits we keep.

FOUND ON THE REAL SYSTEM, THEN FIXED

Approve timed out
Riverpod paused the provider nobody watched

✓ **Fixed: read it once · PR #70**

Dashboard spun offline
count() waited forever without a network

✓ **Fixed: 15 s timeout and an error state · PR #71**

Gemini model retired
new projects got a 404 from the old model name

✓ **Fixed: model name moved to Remote Config**

WHAT WE KEEP DOING

Rules are the backend
Every business rule is a rules condition, so the rules get their own tests (30/30).

Read the brief twice
We removed the PDF, re-read the brief, and put it back: saved on the device.

AI needs a plan B
App Check, quota and model changes broke AI once; writing by hand kept the flow alive.

Conclusion

Nốt is now a working multi-role system, and we can show every claim.

1 Problem solved

One small win in 30 s, Gemini sorts the day, and staff keep the challenges fresh and reach every inbox.

2 Working product

WF-01, WF-02, WF-03 run live on Android and the deployed web admin, on the free plan.

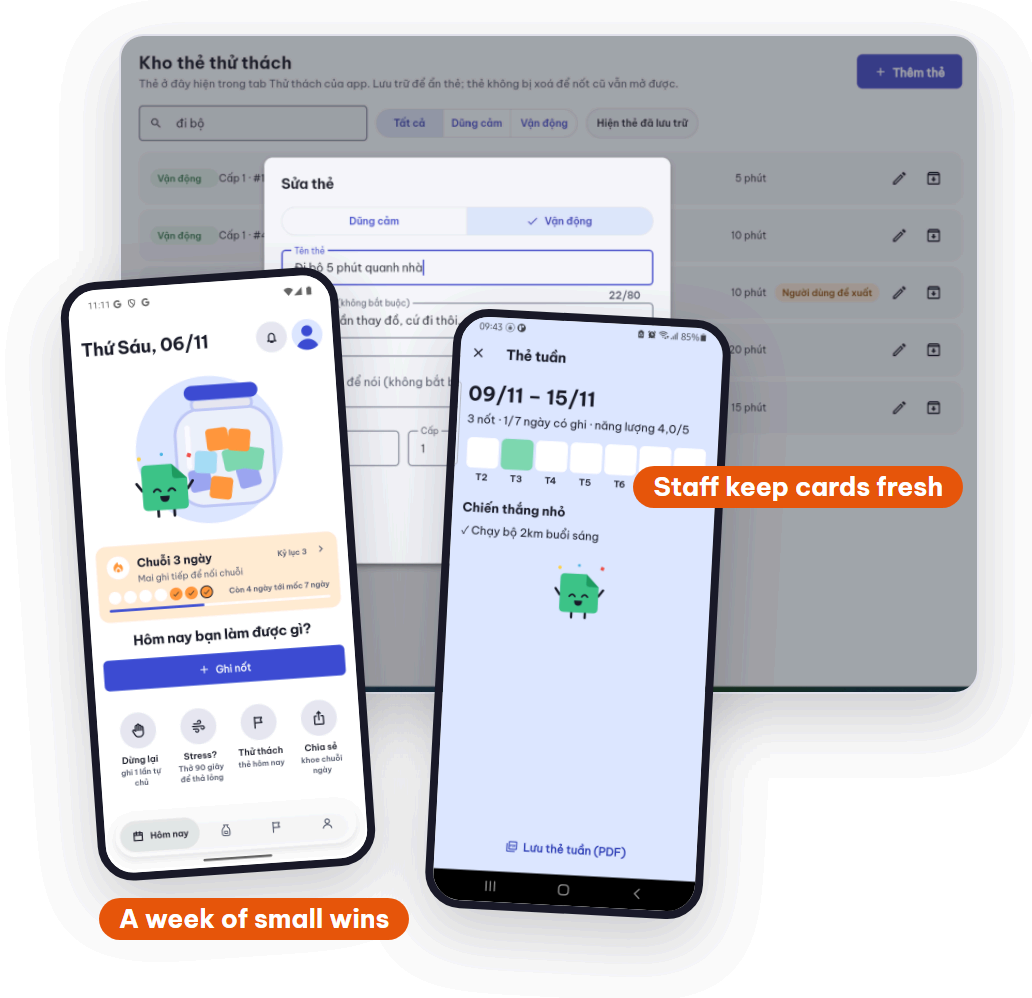
3 Verified

30/30 rules tests · 300+ code tests in CI · 14/15 ARTEMIS runs on a real phone · 3 timed cross-role runs · App Check enforced.

prm393-not-lab3.web.app

/admin/

github.com/FU-PRM-Projects/Lab-03



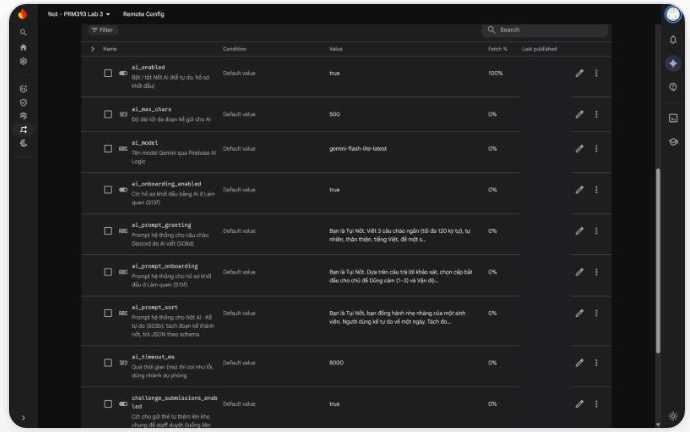
AI examples

Three real Gemini inputs and outputs, and one handled failure.

INPUT (KỂ TỰ DO)	GEMINI OUTPUT
Ran 2 km, ate a snack I meant to skip, calmed down without blaming myself	chien_thang "Chạy bộ 2km quanh hồ" · nhat_ky "Lỡ ăn thêm gói snack..." · tu_chu "Không tự trách và bình tĩnh lại" · mood binh_thuong
Asked a stranger in the library where to charge my phone; wrote a few lines about the rainy afternoon	chien_thang "Lấy hết can đảm hỏi mượn chỗ sạc pin" · nhat_ky "Ghi lại vài dòng cảm xúc buổi chiều mưa" · mood vui
Tired for days, only tidied my desk today	chien_thang "Mình đã dọn dẹp được cái bàn học" · 2× nhat_ky · mood met · needs_support false
Handled failure	404 model retired / 401 missing App Check → `ai_error` logged, paragraph kept, "Ghi tay" offered; fixed by Remote Config model name

```
29
30  GenerativeModel _model(String instructions, Schema schema) =>
31      FirebaseAI.googleAI().generativeModel(
32        model: config.model,
33        systemInstruction: Content.system(instructions),
34        generationConfig: GenerationConfig(
35          responseMimeType: 'application/json',
36          responseSchema: schema,
37          temperature: 0.4,
38        ),
39      );
```

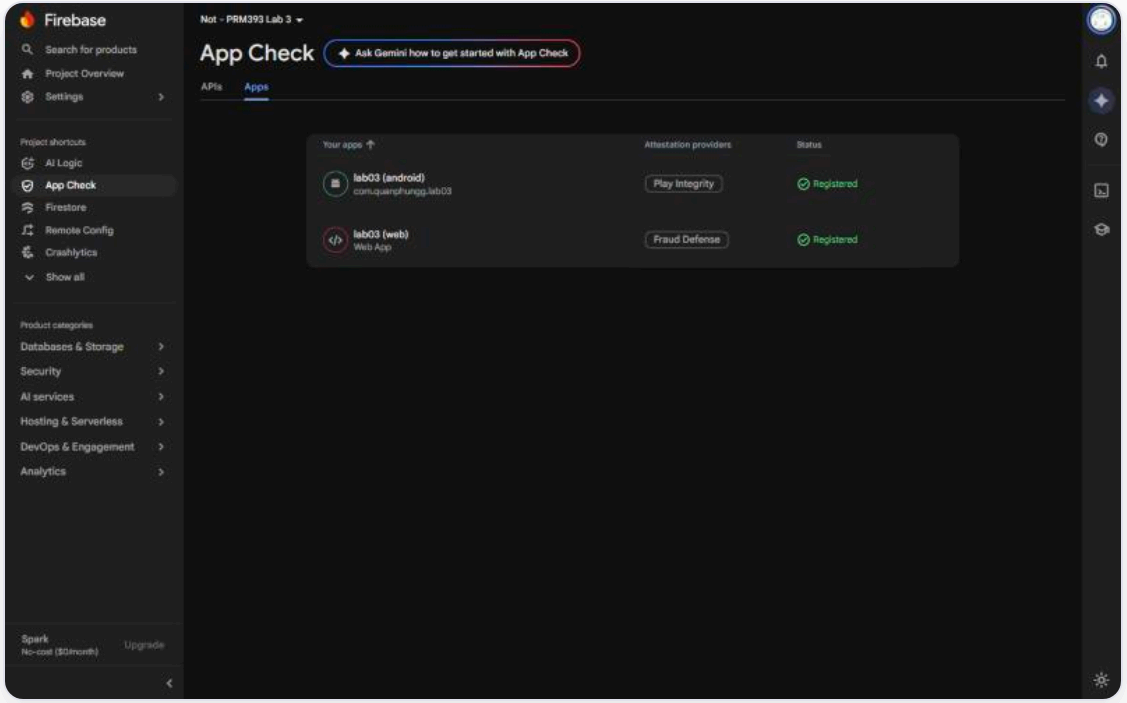
JSON schema forces the answer shape



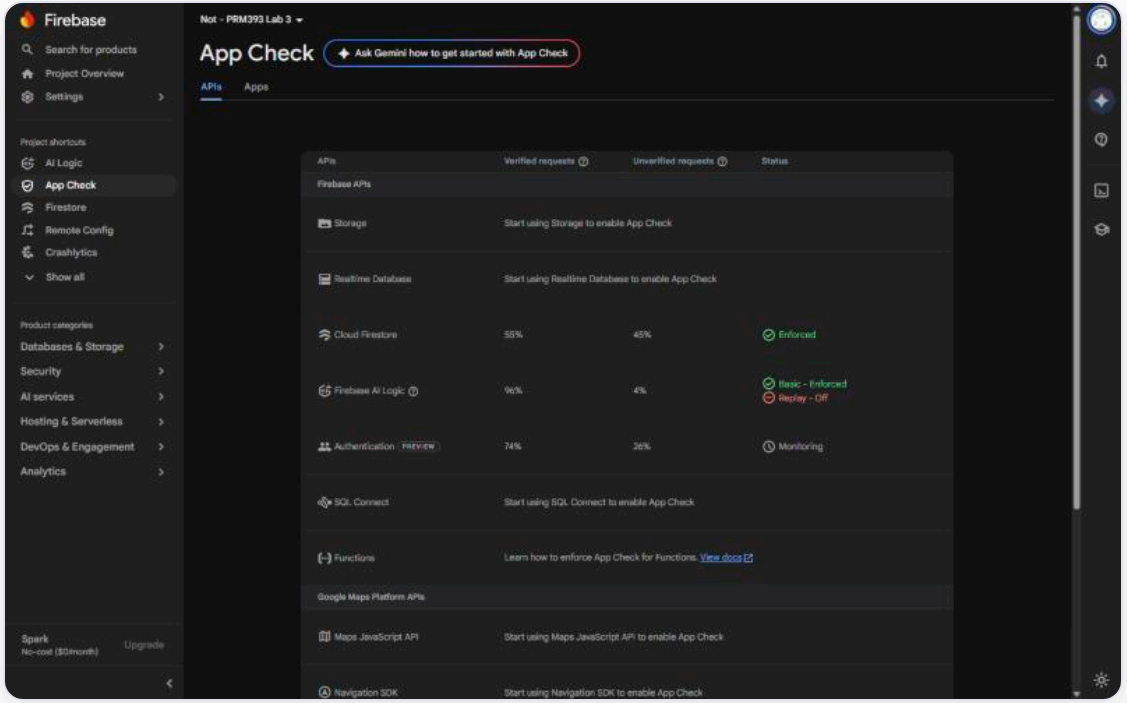
Remote Config: model, prompts, on/off

App Check: registered and enforced

Both apps are registered; Firestore and AI Logic are enforced.



Apps: lab03 (android) Play Integrity · lab03 (web) Fraud Defense



Cloud Firestore: Enforced · AI Logic: Enforced

NỐT · PRM393 FINAL ASSIGNMENT

Thank you

Questions?

prm393-not-lab3.web.app · /admin/ · github.com/FU-PRM-Projects/Lab-03

